THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE TERMINAL EXAMINATION
033 BIOLOGY

MARKING SCHEME MAY 2025

SECTION A (15 MARKS)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
A	C	D	D	A	D	B	B	B	C

1@ = 10

2.

1@ = 05

LIST A	i	ii	iii	iv	v
LIAST B	H	A	D	G	C

SECTION B (70 MARKS)

3.

Rat's cells	Onion's cells
(i) no cell wall	Has cellulose cell wall
(ii) Has centrioles	Has no centrioles
(iii) Has no chloroplast	Has chloroplast
(iv) Has small temporary vacuoles	Has large permanent central vacuole
(v) Irregular shaped	Regular shape
(vi) Stores glycogen	Stores starch

(Any five 02@ = 10 mark)

4. Characteristics of living things
   i. Nutrition: nutrition involves obtaining and using nutrients (food substances) for energy production and growth and repair. Plants manufacture their own food by using simple inorganic substances while animals feed on already manufactured, complex organic substances.
   ii. Respiration: respiration involves the step by step breaking down of food substances to produce energy. Aerobic respiration requires oxygen while anaerobic respiration does not require oxygen.
   iii. Reproduction: reproduction involves the formation of new members of a species from the parents.
   iv. Irritability or sensitivity: irritability refers to response to stimulus.
   v. Movement: movement refers to change in posture or location by an organism. Movement that involves the change in location of the whole body is called locomotion. Locomotion is exhibited mostly by animals, while plants exhibit growth movements.
   vi. Growth: growth refers to the increase in the size of organisms.

vii. Excretion: excretion is the process of removing metabolic wastes from the body. Metabolic wastes are those resulting from various metabolic activities such as respiration and digestion. *Any five 02@ = 10 marks*

5.

(a) Photosynthesis is the process whereby green plants make carbohydrate by using water, carbon dioxide and the sunlight. *02½ marks*

(b)

	Description	Function
Epidermis	The outer layer of the cells covering the,it is covered with a waxy cuticle *02½ marks*	The functions of the epidermis are: protection against water loss by way of transpiration, regulation of gas exchange and secretion of metabolic compounds.
Stomata	Pores in the epidermis, The stomatal pores are surrounded on each side by chloroplast-containing guard cells. In a typical leaf, the stomata are more numerous over the abaxial (lower) epidermis. *02½ marks*	Regulate the exchange of gases and water vapor between the outside air and the interior of the leaf during photosynthesis and respiration. *02½ marks*

6. (a) The cell theory states that:

- All living things are composed of cells
- The cell is the basic unit in living things
- All cells are produced from other cells *04 marks*
- Cells contain inherited information which controls their activities
- All life processes take place in the cell

(b)

Nucleus: to control the functions of all other parts of the cell, - to determine the hereditary characteristics

Cell membrane: to protect the cell and to control the movement of material in and out of the cell. The cell membrane together with the cytoplasm is called protoplasm.

Mitochondria: to produce energy (it is the power house of the cell).

Ribosomes: protein synthesis

Vacuoles: secrete and excrete wastes from the cell.

Chloroplast: photosynthesis. *01@ = 06 marks*

7.

(i) Only the first letter of the generic name is capitalized and then the whole generic name is underlined or italicized; example: Homo. 02 marks

(ii) The first letter of the species name is never capitalized, and then the whole species name is underlined; example: sapiens. So the scientific name for human is Homo – genus, sapiens – species. 02 marks

(iii)If the scientific name is written in italics, then it should never be underlined, but other rules remain the same. Example Homo sapiens or *Homo sapiens* 02 marks

(b)

Disadvantages of Artificial classification

(i) It is not accurate enough as it uses a few observable features. 01 marks
(ii) It is biased; *explain*…. 01 marks
(iii)Ends up with unrelated organisms being put in the same group or vice versa. 01 marks
(iv)It may cause confusion, *explain*.. 01 marks

8.

(i) Studying biology helps us understand ourselves better and therefore maintain our wellbeing by keeping our bodies under optimum conditions.
(ii) Studying biology enables us understanding better other organisms that we interact with, therefore being able to live with them in a way that would benefit us. For example, how to keep benefiting from plants in terms of food source, medicine and environment protection.
(iii)Studying biology provides us with answers to questions about the nature of life, how and where did it begin.
Any five 02@ = 10 marks
(iv)Knowledge of biology helps us in preventing and treating diseases so that we can stay healthy.
(v) Biology helps us to attain careers such as medicine and genetic engineering.
(vi)Studying biology provides us with scientific skills that can enable us to conduct researches

9. Relationship between biology and other science fields

(i) Medicine: Medicine deals with treatment of diseases and injuries. The individuals who are involved in the application of medicine include medical doctors, nurses and veterinarians. A veterinarian deals with taking care of the health of animals. All people who practice medicine need the knowledge of biology.
(ii) Pharmacy: Pharmacy deals with the preparation, handling and selling medical drugs (medicine). A person whose profession is pharmacy is called a pharmacist and the knowledge of biology is required in order to become a pharmacist.
(iii)Agriculture: Agriculture is the field which deals with the production of farm crops and livestock. Since plants and animals are living organisms, knowledge of biology is evidently required in this field for better farm crop production and livestock keeping.
(iv)Nutrition: Nutrition is the science of food and its health effect in the body of an organism. A person whose profession is nutrition is called a nutritionist. Knowledge of biology plays major part in the field of nutrition.
(v) Forestry: Forestry deals with planting and taking care of trees and forests. A person whose profession is forestry is called a forester. Studying biology is an essential part of the forestry profession.

Any five 02 @ = 10 marks

SECTION C (15 MARKS)

10. Introduction: Definition of biology *01 marks*

Biology is divided in to two main branches, Botany and Zoology. **Botany** is the scientific study of plants while **Zoology** is the scientific study of animals. *02 @ = 04 marks*

Other branches of biology are:

1. Anatomy: the study of the structure of plant and animal organs.
2. Histology: the study of the structure of tissues.
3. Microbiology: the study of microscopic organisms.
4. Bacteriology: the Study of bacteria.
5. Virology: the study of viruses.
6. Mycology: the study of fungi.
7. Cytology: the study of the structure and function of cells.
8. Ecology: the study of organisms and their interaction with each other and the environment.
9. Entomology: the study of insects.
10. Parasitology: the study of organisms which live and feed on/in the bodies of other organisms.

Any six 01½ @ = 09 marks

Conclusion: Any relevant *01 marks*

(Total: 15 marks)